

Temperature probe

DETECT

DS18B20

DIGITAL

What does it do?

A **DS18B20** measures temperature and sends a digital reading in degrees Celsius.

This probe has a sealed metal tip on a cable with three wires. Check the supplier's pinout; wire colours can vary.

Only the sealed probe goes in water. Keep the board and all wire connections dry.

Uses



Aquarium temperature alarm

DETECT
Measure tank temperature

DECIDE
Is it too cold?

DO
Show a warning



Penguin bath alarm

DETECT
Measure bath temperature

DECIDE
Is it too warm?

DO
Light a warning LED



VDD
Power
from 3V3

DQ
Digital
data

GND
Ground

4.7 kΩ
Resistor
from DQ
to 3V3

Code

● Raspberry Pi

```
from w1thermsensor import W1ThermSensor

sensor = W1ThermSensor()
temperature = sensor.get_temperature()
print(temperature)
```

● Raspberry Pi Pico

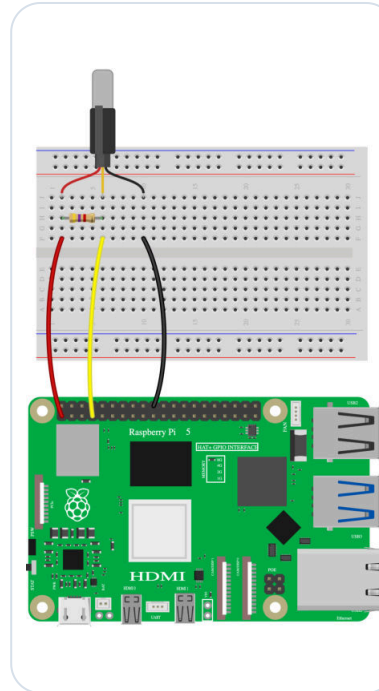
```
from machine import Pin
import onewire, ds18x20
from time import sleep_ms

bus = onewire.OneWire(Pin(28))
sensor = ds18x20.DS18X20(bus)
probes = sensor.scan()
assert probes, "Check probe wiring"
sensor.convert_temp()
sleep_ms(750)
temperature = sensor.read_temp(probes[0])
print(temperature)
```

Pi setup: enable 1-Wire in [raspi-config](#), reboot, then run [sudo apt install python3-w1thermsensor](#). Pico: repeat the conversion and 750 ms wait for each new reading.

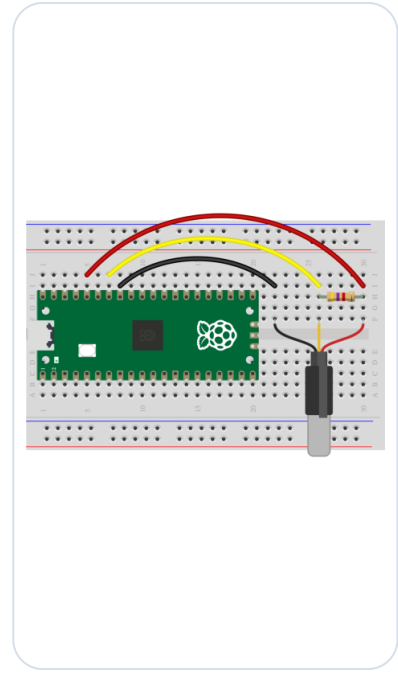
Wiring

RASPBERRY PI 5



VDD → 3V3 (1); DQ → GPIO4 (7); GND → GND (20). 4.7 kΩ between DQ and 3V3.

PICO



VDD → 3V3 OUT (36); DQ → GP28 (34); GND → GND (33). 4.7 kΩ between DQ and 3V3.